

Project Initialization and Planning Phase

Date	25 June 2026
Team ID	
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

Section	Content
Project Overview	Emotion Detection & Learning Support Engine is an AI-powered educational platform that detects emotions from student text and provides personalized learning guidance using Machine Learning, NLP and Generative AI.
Objective	Develop an intelligent assistant capable of detecting student emotions and generating adaptive academic guidance.
Scope	The system analyzes text, classifies emotions, generates AI responses, displays analytics, and supports future cloud deployment and multilingual learning.

Topic	Details
Problem Statement	Students often receive generic learning support that ignores their emotional state, reducing engagement and learning effectiveness.
Impact	Lower motivation, reduced confidence, poor academic performance, and inefficient self-paced learning.
Proposed Solution	Use Machine Learning and NLP to detect emotions from text, then generate personalized explanations and study guidance using a Large Language Model (Groq Llama 3.1 API).
Key Features	- Emotion Detection - Personalized Learning Suggestions - AI-generated Responses - Emotion Analytics Dashboard - Adaptive Learning Recommendations

Resource Requirements

Resource Type	Description	Specification
Hardware	Processor / RAM / Storage	Intel i3/i5+, 4 GB RAM, 100 GB Storage
Software	Development Tools	Python 3.10+, VS Code, Streamlit, Git, Scikit-Learn, Pa
Data	Training Dataset	Labeled emotion text dataset (CSV)